

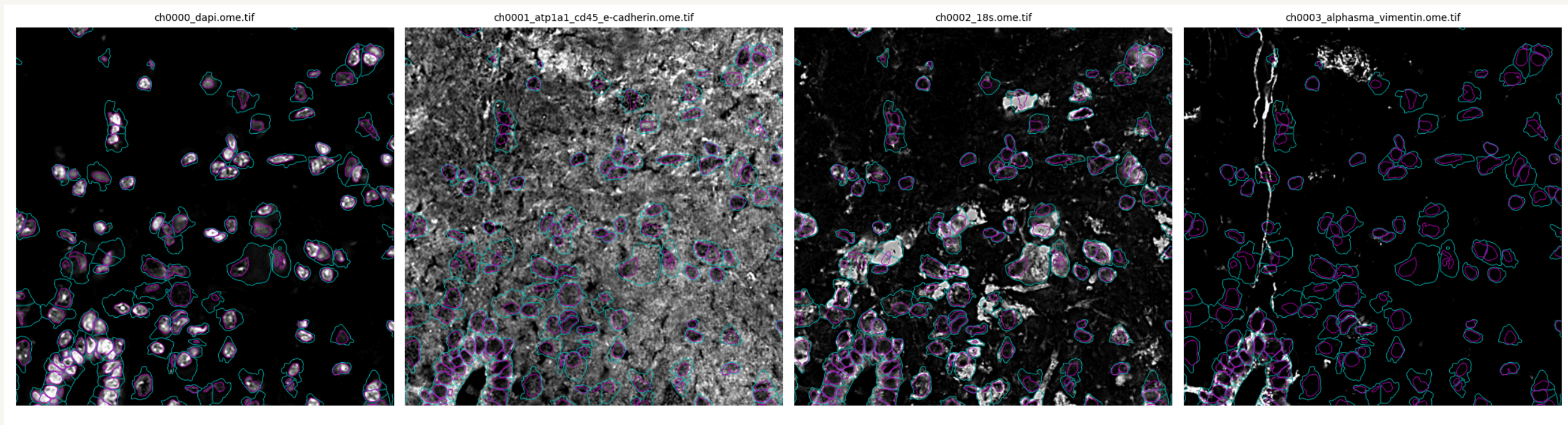
■ BEYOND BOUNDARIES · LAB MEETING · 7 OCT 2026

What does the Xenium *multimodal segmentation kit* tell us that the transcriptome doesn't?

Mouse EAE spinal cord, RRMAP2 runs 5 + 6 · 18 Xenium 5K images · 51 tissue pieces · 25 animals · 500,379 annotated cells · 201 image features per cell

THE QUESTION

Four stains are imaged anyway, to draw the cell masks. *Is there biology in them?*



DAPI · ATP1A1/CD45/E-Cad (boundary) · 18S rRNA (interior; drew ~95 % of masks) · αSMA/Vimentin. Per cell: intensity in nucleus / cytoplasm / rim / surrounding ring, texture, polarity, shape. **Vim, Acta2 and Atp1a1 are not on the 5K panel.**

●
■ BOTTOM LINE

Three kinds of added information. *No hidden cell states.*

Protein

...for genes not on the panel

Vimentin, αSMA, ATP1A1 per cell. Astrocyte vimentin lags reactive RNA in active disease.

Tissue

...context around each cell

Neuropil loss at the lesion's grey/white-matter border (~15–20 %; the first ~25 % was mostly distance to grey matter). Anatomy that adds to the transcriptome (0.68 → 0.73).

Images alone

...recognise cells and map lesions

Cell type 46 % (chance 7 %), lesion maps AUROC 0.90, and this holds **across imaging runs**.

● Not added: CD45 (undetectable in mouse cord); lesion identity and clinical score beyond what the transcriptome already gives. ~85 % of image variance is unexplained by RNA, but that's mostly optics.

DO THE STAINS MEASURE WHAT THEY CLAIM?

Known biology: yes, in every animal. *CD45: no.*

0.86

ependymal vimentin

0.78

neuronal 18S (cytoplasm)

0.73

leukocyte vimentin >
neuron / oligo

0.66

VSMC αSMA-bright

0.48

leukocyte CD45 at the rim:
chance; boundary channel
= ATP1A1 neuropil only

AUROC; positives > 0.5 in 25/25 animals; also in boundary-segmented cells. → Ask 10x whether the kit's CD45 antibody is mouse-reactive.

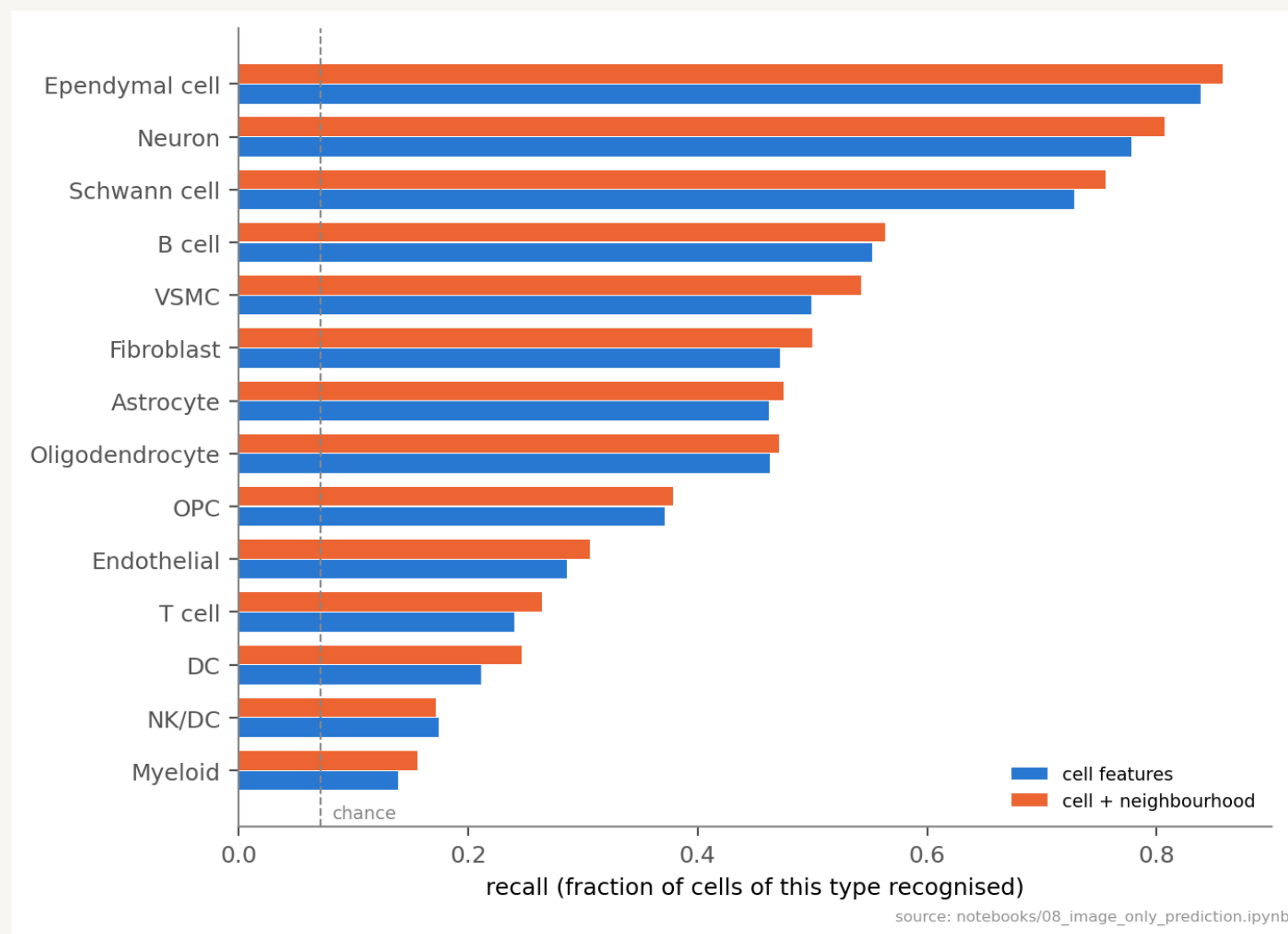
■ HOW MUCH OF THE IMAGE IS NEW?

Most of the image isn't in the transcriptome, *but most of that is optics*

- Covariates + 50 transcriptome PCs explain a median **~15 %** of each image feature (best: nucleus:cell ratio, rim/cyto intensity).
- The remainder is spatially coherent, but largely a **local field shared across cell types** ($r \approx 0.38$; follows neuropil glow, not lesions).
- The cell-specific remainder barely moves with lesion state: 19/924 tests, ~ 0.1 SD.
- Residual clustering: no discrete hidden states, only continuous gradients.

The kit is not a source of new *cell types*. It's a source of new *measurements* for specific questions.

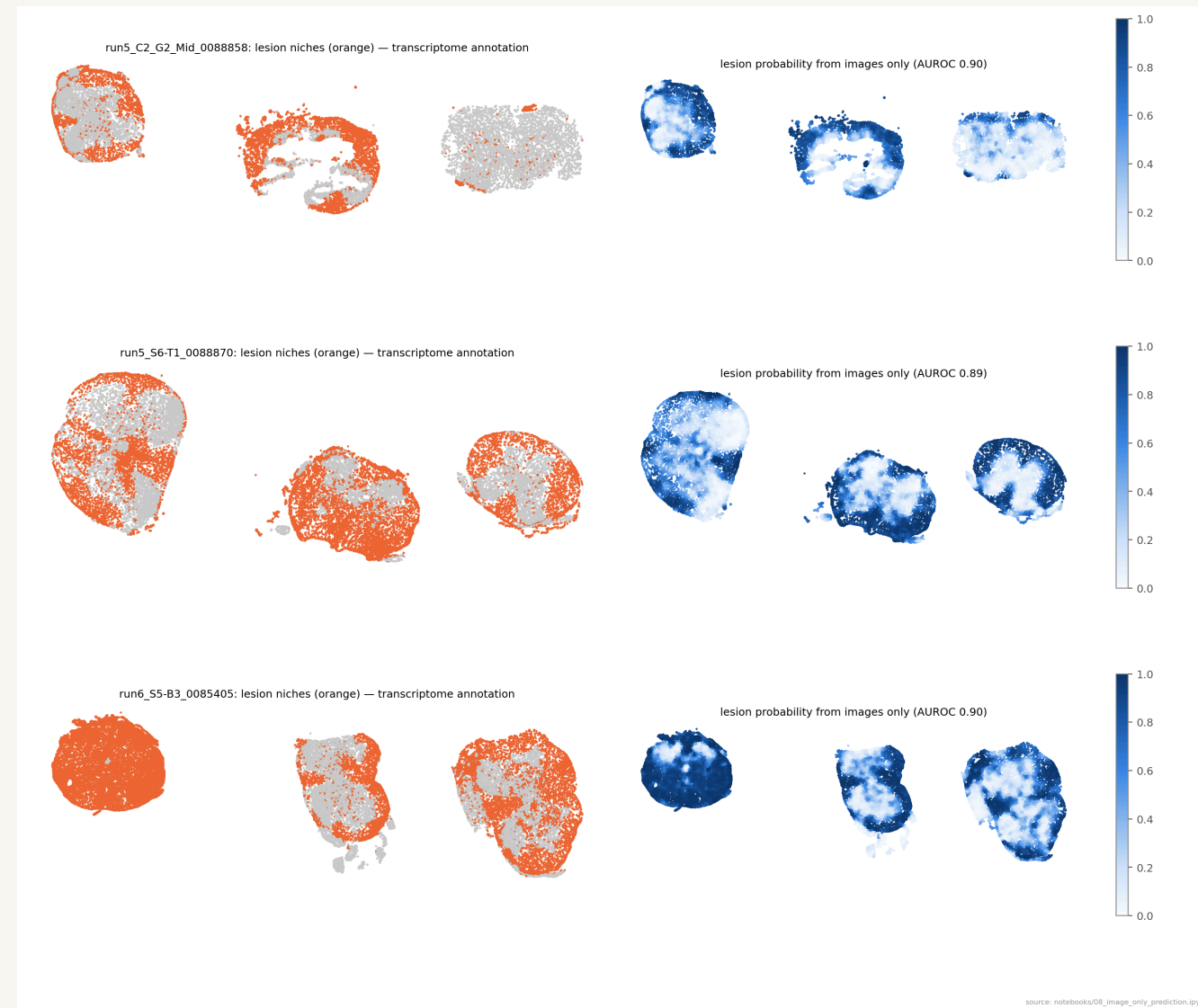
IMAGES ALONE, HELD-OUT ANIMALS



No transcripts: *46 %* of 14 cell types
(chance 7 %)

- Ependymal 0.86 · neuron 0.81 · Schwann 0.76
- Immune poor (myeloid 0.16, T 0.26): no CD45
- Errors stay within lineage
- Subtypes: microglia 72 %, reactive astrocytes 68 %, cholinergic neurons 72 %
- Identity is spread across all four stains, so it's not an artefact of the 18S masks

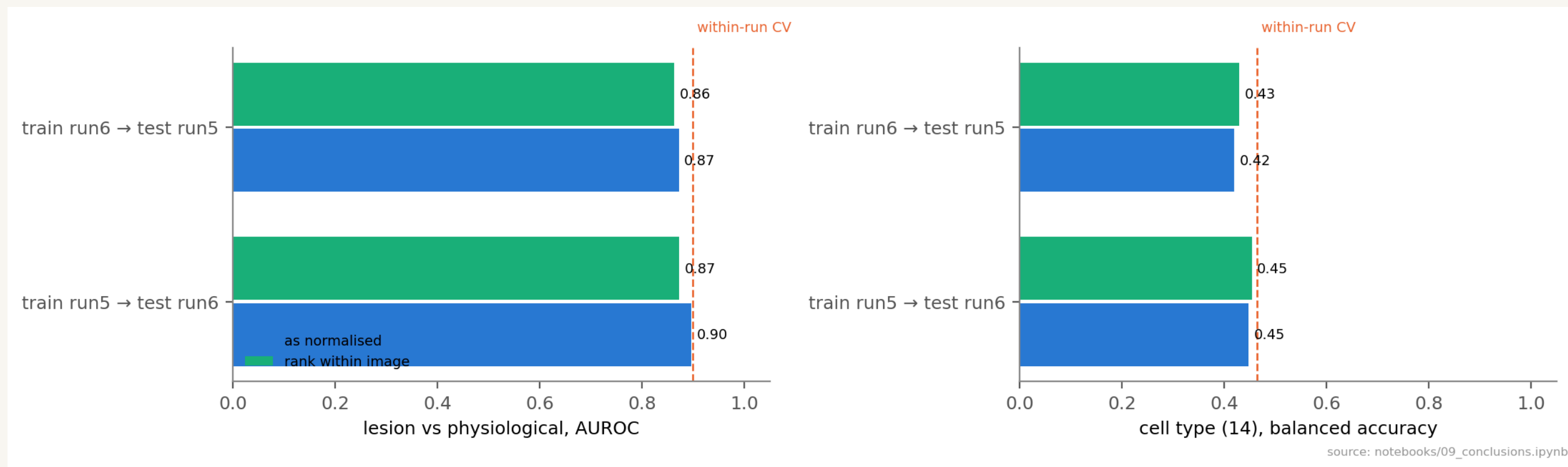
Lesion maps from images: *AUROC 0.90* (transcriptome neighbourhood 0.92)



Top signal: vimentin-channel texture/intensity in surrounding tissue. Anatomy (10 regions): images 0.63, transcriptome 0.68, both 0.73.

NEW TODAY · DOES IT SURVIVE A NEW RUN?

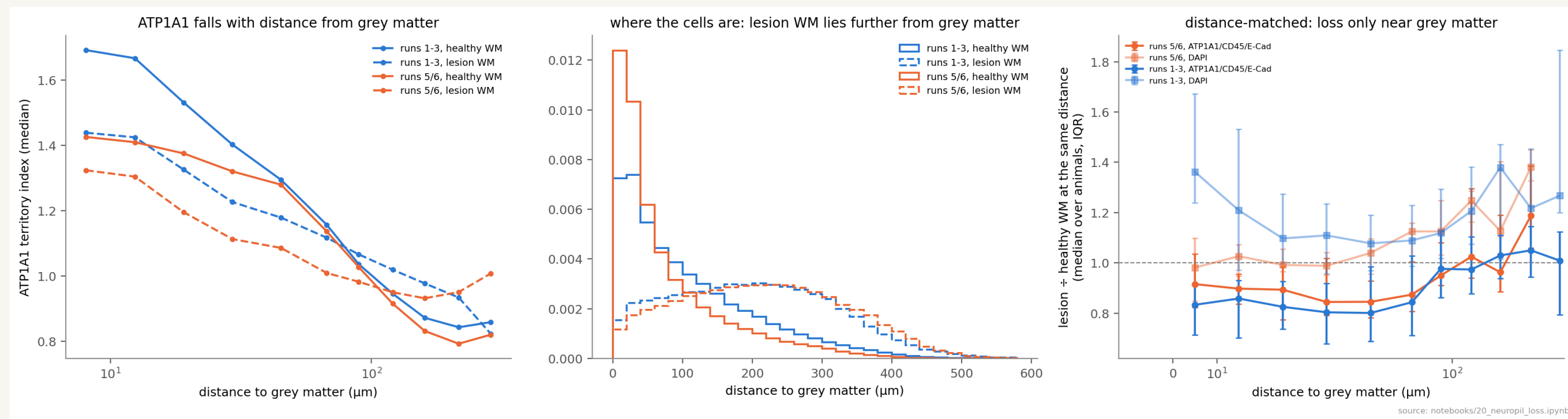
Images identify their run with 100 % accuracy, *yet the models still transfer*



- Train run5 → test run6: lesion AUROC **0.90** (per animal 0.91); run6 → run5 **0.87**. Within-run: 0.90.
- Cell type 0.42–0.45 vs 0.46 within-run. Per-image rank transform doesn't help, so per-image normalisation is enough.

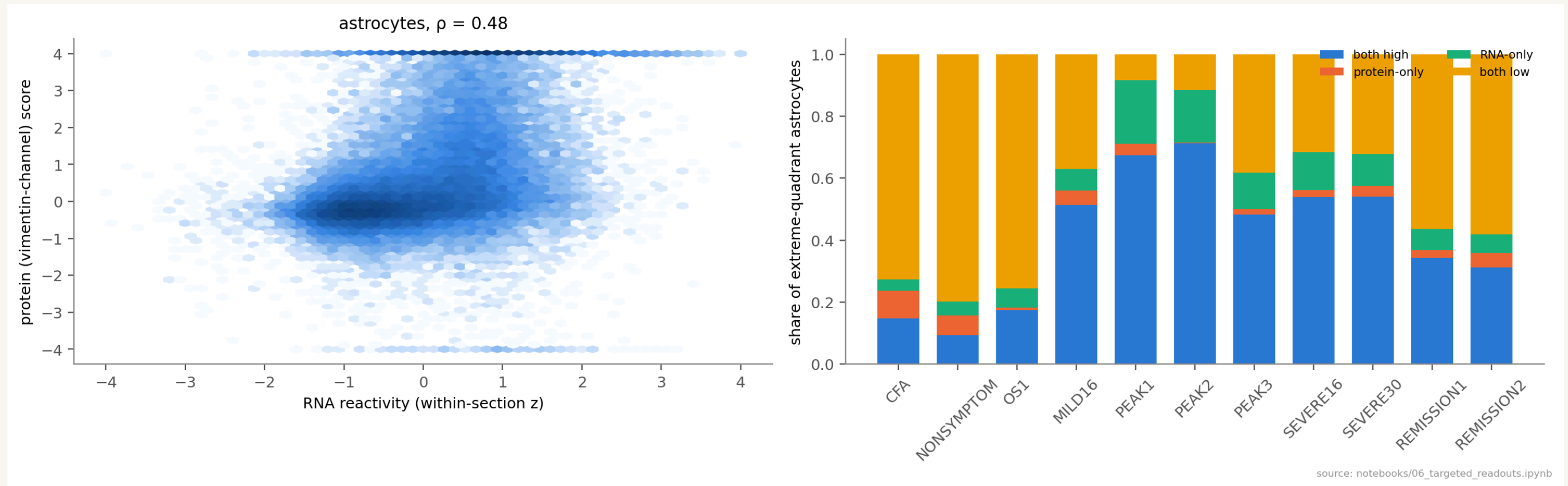
BIOLOGY ONLY THE IMAGES SHOW · 1

Neuropil loss sits at the lesion's *grey/white-matter border*



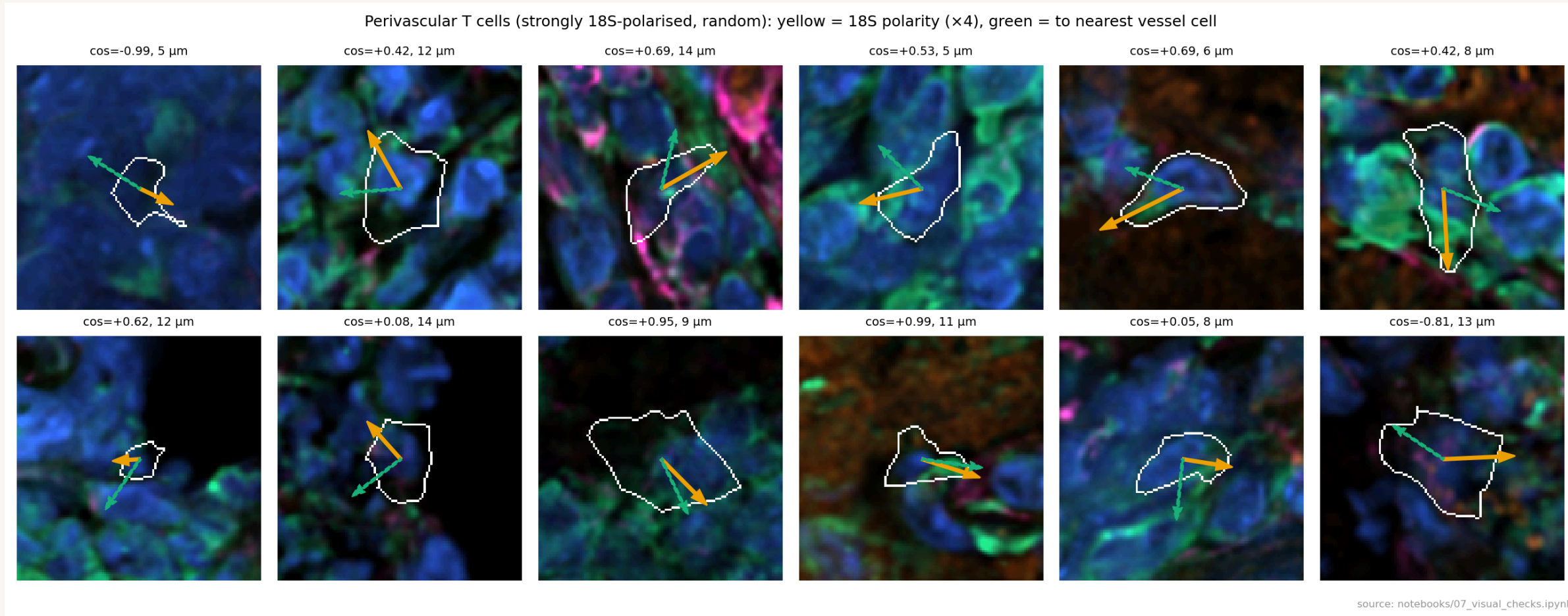
ATP1A1 halves over the first ~150 μm away from grey matter, and lesions sit deeper in white matter. **At the same distance:** ×0.80–0.92 within ~75 μm of grey matter in both run sets (×0.78, 25/31 animals; ×0.83, 13/18), **no loss deeper**. The first "–25 %" was mostly this gradient. Neuron-derived RNA between cells drops at the same place (notebook 23).

● Astrocyte reactive RNA *precedes* vimentin protein



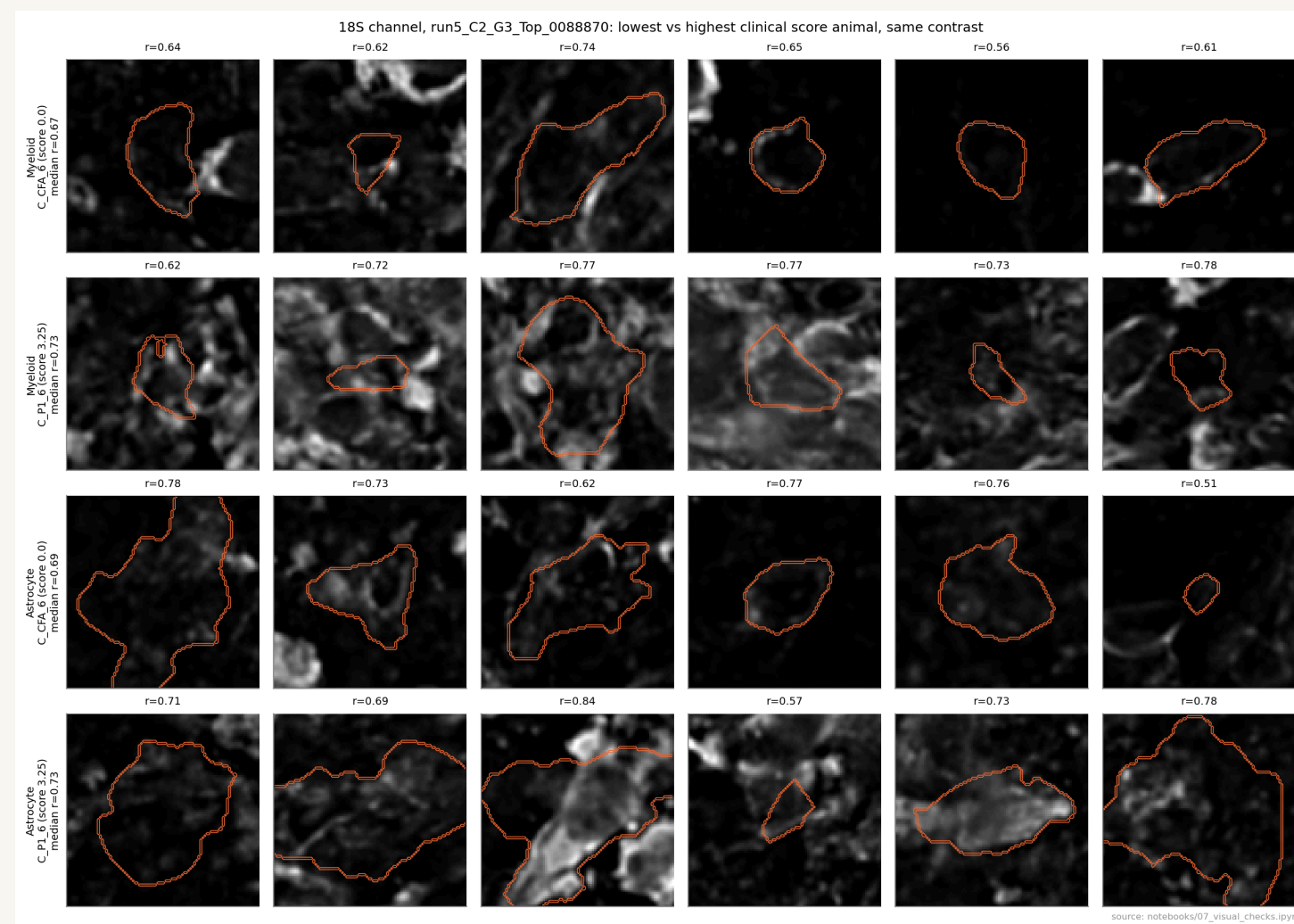
- Protein vs RNA reactivity $\rho = 0.48$ (C3 0.50, Gfap 0.38; Vim itself isn't on the panel)
- **RNA-only** astrocytes peak in active disease (PEAK1 20 %, CFA 3.5 %), fall in remission (active vs remission $p = 0.019$)

18S rRNA: activation (*T-cell orientation withdrawn*)



- **Withdrawn:** the apparent "T cells point 18S at vessels" is bleed from 18S-bright neighbours plus outline geometry (brightness-matched control, notebook 19)
- Lesion 18S rises beyond RNA content in activated cells (Schwann +0.8, endothelium +0.4 SD); the oligodendrocyte drop is just RNA loss

CANDIDATE · NEEDS REPLICATION



Smoother 18S texture in *sicker* animals

- Rises with clinical score in 6 cell types
- Holds **within section** (p 0.53–0.86) and after a crowding adjustment (0.41–0.81)
- DAPI texture (same optics) doesn't follow, so it's probably not focus
- Ribosome redistribution / RNA degradation, or tissue handling per animal

Animal level: score from images p 0.62 vs transcriptome 0.79. Images are informative but redundant.

TAKE-HOME

How to use the segmentation stains

Use them for

Don't use them for

Vimentin / αSMA / ATP1A1 protein (targets missing from the panel)	CD45 / immune membrane (undetectable in mouse cord)
Tissue context: neuropil loss, anatomy, lesion maps	Replacing the transcriptome for lesion identity or severity
Protein-vs-RNA timing (astrocyte reactivity)	Cell polarity from DAPI or αSMA/Vim (optical bleed)
Per-cell 18S as a ribosome / activation readout (adjusted for transcript density)	Pooling <i>raw</i> feature values across runs (models transfer; raw values don't)
QC: segmentation method, spill-over-prone neighbourhoods	18S distribution features as biology in 18S-segmented cells

CAVEATS & NEXT STEPS

Caveats

- Lesion niches are transcriptome-defined, so image-vs-RNA comparisons for lesions are tilted toward RNA
- α SMA and vimentin share a channel
- 18S drew ~95 % of the masks
- 25 animals, 2 CFA pieces; 2 runs from one batch

Everything: notebooks/00_summary · 09_conclusions · report/BeyondBoundaries_summary.pdf

Next

- **Replicate on runs 1–3** (P drive): 18S texture, WM neuropil loss, astrocyte RNA → protein lag, image-model transfer
- Separate vimentin from α SMA (VSMC-adjacent subset); check CD45 reactivity with 10x
- Feed the neuropil-loss index and per-cell 18S into RRMAP2 lesion/niche analyses as covariates